

# A Single-Exponential Exponent for Arithmetic Progressions

A quantitative proof modification for OpenAI/math 159

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## Abstract

This note quantifies the exponent in the density bound of *Quasipolynomial Bounds for Arithmetic Progressions* (OpenAI/math, September 23, 2026). Relative to that manuscript's analytic framework, we obtain  $\varepsilon_k = \exp(-k^C)$ , for an absolute constant  $C$ . The proof keeps a polynomial geometric degree and replaces two sequential inverse-theorem chains by degree descents. A direct Cauchy–Schwarz calculation integrates supplied derivative correlators, while effective nilpotent cube completion removes one vertex in relative detection. Separating structural dimension from logarithmic precision controls the remaining branching arguments. The geometric and analytic modifications and the final numerical schedule are given explicitly, relative to the source results listed below.

## 1 The statement and the scope of the proof

For integers  $k \geq 3$ , let  $r_k(N)$  be the largest cardinality of a subset of  $\{1, \dots, N\}$  containing no nonconstant  $k$ -term arithmetic progression. All logarithms are natural. Write  $\mathcal{M}$  for the 198-page manuscript [1]; its section and page numbers below refer to that version.

**Theorem 1.1** (Quantitative consequence of the source framework). *Subject to the analytic results of  $\mathcal{M}$  specified below, there is an absolute constant  $C > 0$  such that for every  $k \geq 3$  there are constants  $K_k, c_k > 0$  satisfying*

$$r_k(N) \leq K_k N \exp[-c_k(\log N)^{\varepsilon_k}], \quad \varepsilon_k = \exp(-k^C), \quad N \geq 3. \quad (1)$$

**Imported results.** We use the bottom absolute-patch theorem ( $\mathcal{M}$ , Lemma 2.2), the sampler and rank-transfer contracts (Sections 3–4 and Appendices F–H), the comparison and exact-marking framework (Sections 5–8 and Appendices A–E and I), the extraction of Section 9, and the density-increment construction and parameter order of Section 10. We retain their analytic conclusions, including the prescribed dependencies of warm bounds. The algebraic modifications and the dependence of the numerical powers on the fixed orders are the subjects of this note. This is a proof relative to that manuscript, not a replacement proof or an independent verification of all its analytic arguments.

The external inverse input is the proof of Leng–Sah–Sawhney [2], using Leng's equidistribution result [3]. Its published fixed-order notation does not by itself specify the order-dependence needed here; Section 4 records the additional budget calculation. The supplied-derivative integration statement needed for the improvement is proved in Proposition 4.5.

**Budget conventions.** A growing budget contains dimensions, logarithmic rational heights, logarithmic Lipschitz bounds, logarithmic inverse errors and correlations, and logarithmic scale requirements. Order-dependent positive factors  $K_q$  may be arbitrarily large: they are coefficients, not powers of the growing budget. We retain every power of that budget and every successive substitution of one such bound into another. Every exponent implicit in  $O(1)$  below is absolute. For example, a map  $x \mapsto K_q(2+x)^b$ , applied successively  $m$  times, has sufficient degree  $b^m$ ; calling  $m$  fixed for a fixed  $q$  does not justify dropping it from an estimate in  $q$ .

Set

$$s = k - 2, \quad D = 256k^2, \quad p = 2 + \log(1/\alpha)$$

for a starting density  $\alpha$ . The original multiplicative density gain requires  $T \leq K_k p$  rounds. At weight  $h$ , the determining data consist of integer coordinates  $b_h$ , a rational value space  $W_h$ , and targets

$$C_h(u, b_{<h}) \in W_h, \quad \deg_{\text{wt}} C_h \leq h.$$

The spatial variable has weight one and  $b_h$  has weight  $h$ . The paired determining cutoffs force a unique padded integer tuple. Writing  $Q_h = \log(2/w_h)$ , a bound is *warm at  $h$*  if it may use dimensions,  $p$ , and already chosen data at weights strictly above  $h$ , but none of the incoming  $Q_i$  for  $i \leq h$ . A preliminary bound at  $h$  may also use  $Q_h$ . Once retained as history, such data are warm at every subsequent layer  $j < h$ .

The principal bounds proved below are

$$D = O(k^2), \quad \max_{r \leq D+2} I_r, \max_{q \leq D} R_q, F, B \leq \exp(D^{O(1)}).$$

Here  $I_r$  is an inverse-theorem budget exponent,  $R_q$  is the relative-detector exponent of Section 5, and  $F$  controls the fresh absolute rank, and  $B$  controls the source's local polynomials and their descending numerical schedule. The last density calculation uses the original rank ledger.

## 2 A persistent record with quadratic degree

A completed active return has two kinds of controlled factors: a slow factor in normalized coordinates and a rational factor in integer coordinates. Keeping these factors on their original sides allows the middle orbit to retain its original degree.

For a filtered Lie group, call a polynomial *jointly filtered* when each monomial of weighted degree  $a$  in its logarithm has coefficient in the  $a$ -th Lie layer. Constants lie in the first layer. Integer and normalized labels have weight one. A *marking* is the fixed projection  $\pi : G \rightarrow Y$  recording the determining equations.

**Lemma 2.1** (Ordered active return). *In the setting of Theorem 8.2 of  $\mathcal{M}$ , suppose the input orbit is jointly filtered in all actual parameters and its marking is a fixed filtered polynomial  $P_Y(X)$  of the site variables. The group and marking may themselves be finite charged families in older dummy variables, which are coefficients rather than actual arguments. The return can be chosen in the form below, with each of  $E, C, R$  jointly filtered in the original filtration:*

$$g^\sharp(X, \theta, \tilde{t}, z) = E(X, \theta, z) C(X, z) R(X, \tilde{t}, z), \quad \pi E = \pi R = 1, \quad \pi C = P_Y(X). \quad (2)$$

Here  $X$  collects the site variables,  $z$  the short labels, and  $\theta, \tilde{t}$  the normalized and integer long labels. The factor  $C$  has no long-label argument;  $E$  has controlled coefficients in normalized site, residual, and label coordinates; and  $R$  has controlled common denominator in raw integer coordinates. No coefficient bound is required for  $C$ . The original score slice, both expected-payoff representative identities, and the field estimates are retained.

The additional induction proving the lemma is given in Section 3. We first show how its format closes the macro argument.

**Lemma 2.2** (Transport of entire residual features). *At successive active layers, a historical normalized real label can be stored as an affine expression in the current normalized spatial coordinates and named higher residuals. At final reassembly every such feature has weighted degree at most  $s$ . Unfrozen historical short integer labels have a finite menu of whole site functions, with preliminary cost at their creation layer and warm cost at every lower layer. These statements concern the exactly marked payoff and retain both of its zero cases. Only active layers create these histories. Old determining slots of weight  $h > s$  remain in the canonical passive site multipliers and are not included among these features.*

*Proof.* Fix one edge at layer  $j$ , and write its parent higher raw target as  $C_h^P(u, b_j, \dots, b_{h-1})$ , where  $j < h \leq s$ ; lower variables already fixed at this parent have been absorbed into  $u$ . The edge has an affine spatial map  $\psi(t)$  and an exact integer current lift  $B_j^\psi(t) = \beta_j^\psi(t) + b_{j,0}$ . By the definition of the stopped tree in  $\mathcal{M}$ , Section 3.5, p. 23, the child targets are defined by the formal identity

$$C_h^\psi(t, b_{j+1}, \dots, b_{h-1}) = C_h^P(\psi(t), B_j^\psi(t), b_{j+1}, \dots, b_{h-1}). \quad (3)$$

Undo any fixed triangular determining reindexings on both sides, using their recorded exact inverses. For the named raw residuals  $\rho_h = b_h - C_h - \ell_h$ , it follows that

$$\rho_h^\psi(t, b_{>j}) = \rho_h^P(\psi(t), B_j^\psi(t), b_{>j}). \quad (4)$$

The centered residual used by the sampler is an affine translate and sign change of this feature:  $y_h = -\rho_h + (b_{h,0} - c_h - \ell_h)$ .

Use the fixed prepared offsets  $\ell_h$  in these residuals; newly selected smaller-box centers remain cutoff parameters. The canonical tuple is defined on one fixed prepared padded chart: take its true current raw lift, then the successive higher padded lifts, using the prepared offsets rather than path-dependent new centers. Outside that chart use the common zero extension. In the next external field, replace an old residual *as an entire feature* by its parent version evaluated at this canonical tuple. On actual paths this agrees by (4). On an exactly marked return, the determining transformation is the parent transformation, and a contributing determining cutoff forces that same canonical tuple. Keep the whole integer tuple sum. Each inadmissible tuple contributes zero to both expressions; for an admissible tuple all substituted features agree. If no padded tuple exists both payoffs are zero. This proves separate equality for the two unsigned payoffs. Off the scoring locus this is a definition of the external field, just as in  $\mathcal{M}$ , Lemma 7.5; no claim that rounding is a polynomial is made.

During the next micro return, the previous spatial coordinates are parameter labels. After the current macro extraction and freezing of nonpivot parameters,  $\mathcal{M}$ , (9.4), expresses these normalized real labels affinely in  $u/H$ ,  $y_j/v_j$ , and the normalized active sheet coordinates. The latter are linear in the same residual frame. The normalized extra inactive labels are also affine in that frame after their product columns have been fixed, by (9.2). Update only these affine spatial features, and transport the already named higher residuals by (4). This proves the affine feature invariant by induction. At final reassembly insert  $\rho_h = b_h - C_h(u, b_{<h}) - \ell_h$ ; it has weighted degree at most  $h \leq s$ . In particular, one never composes an arbitrary degree- $h$  child target directly with a degree- $j$  pivot, which could have degree  $hj$ .

For the integer menu, an unfrozen extra short label created at  $h$  has the form  $w(u) = \lambda \beta_h^{\text{can}}(u) + c$ , with integral  $\lambda, c$ . Lemma 3.1 and (3.1) of  $\mathcal{M}$  give

$$\|\lambda\| \leq \exp(\text{poly}(d_h)) K_i, \quad K_i \leq k_i/v_h.$$

For a short axis,  $k_i$ ,  $v_h^{-1}$ , and the product-column offset have preliminary bounds. Keep the row in the original raw coordinates by undoing representation reindexings. If one writes  $\beta_h = b_h - b_{h,0}$ , the additional offset is controlled as well:  $\lambda$  annihilates  $W_h$ ,  $c_h \in W_h$ , and the feasible-center relation (3.6) controls  $b_{h,0} - c_h - \ell_h$ . For  $m_h$  labels and  $\|\lambda\|, |c| \leq R_h$ , there are at most

$$(2R_h + 1)^{m_h(d_h+1)}$$

row-and-offset lists. Their logarithmic count is polynomial in the structural dimensions and the preliminary budget. These lists specify *whole functions* at the fixed later parent. No regularity or coefficient bound on the canonical site map is used. Annihilation of  $W_h$  also bounds the values throughout its padded chart. All these costs are warm at  $j < h$ .

For physical reassembly replace canonical integers only in these external features by the formal summation integers. The equality just proved holds on the full chart and exact-inactive scoring locus. This is the locus on which the two representative identities are required in (9.8) of  $\mathcal{M}$ ; apply them before freezing stochastic constants.  $\square$

**Proposition 2.3** (Persistent macro record). *Starting from an absolute patch of degree  $s$  and rank  $d_0$ , the macro argument can retain the original  $d_0$  absolute slots and a middle orbit of joint degree at most  $s$ . Its final physical formula has absolute-weight ceiling at most  $s(s+1)$ . The representation, field, score, and extraction contracts of  $\mathcal{M}$  are preserved, with warm costs at each later layer. The padded ceiling  $D = 256k^2$  therefore suffices.*

*Proof. The retained data.* Store the return as  $E_{\text{hist}}CR_{\text{hist}}$ , and use only  $C$  as the next active orbit. Historical integer short labels remain formal coefficient variables, with *one total-charge cap  $s$*  over all their blocks. Raw and normalized copies of a short label always use the same variable and its fixed normalizing divisor. Historical long labels remain in the outer factors. A deep coefficient lattice evaluates into the pointwise lattice at every integer short label; arbitrary real labels have no such property. Before the next positive-filtration recursion, normalize any degree-zero family constant as in the proof of Lemma 2.1.

A newly promoted determining coordinate retains its original weight in  $C$ . In outer-factor coefficients use its canonical site value: the left factor uses the bounded normalized residual frame, and the right factor uses raw integer coordinates. Exact marking identifies these values at every tested point. Fix the integral determining reindexings before recording history, and keep their exact triangular inverses. They preserve the original determining degrees. Later congruence restrictions have their creation-layer costs, hence warm costs at every lower layer.

*Ordered representation.* Normalize the middle factor only. For an integral triangular reindexing  $T$  of the absolute tuple, the correct expression is

$$E_{\text{hist}}(CT)(T^{-1}R_{\text{hist}}T). \quad (5)$$

This is the old physical map followed by  $T$ , which permutes the complete integer tuple sum. Conjugation by the integral map preserves controlled rational denominators; its numerator need not be bounded. Coefficientwise reduction gives  $CT = Sg$ , where  $S$  is the bounded full-weight map. It is independent of the historical variables: their positive degree excludes them from a full-weight slot term. If  $g = E_{\text{new}}C_{\text{new}}R_{\text{new}}$ , collect the result as

$$(E_{\text{hist}} \text{Ad}_S E_{\text{new}})(SC_{\text{new}})(R_{\text{new}}R'_{\text{hist}}), \quad R'_{\text{hist}} = T^{-1}R_{\text{hist}}T. \quad (6)$$

Only bounded maps conjugate slow factors. Rational right factors stay on the right. The triangular subgroup in which determining transformations have no absolute arguments is bracket closed; forgetting the absolute slots is strong, since a determining shear lifts while fixing them. Thus this normalization preserves the exact determining projection.

*The quotient fields.* At site  $u$ , define the field on the coefficient-family quotient by the original cutoff sum evaluated on

$$E_{\text{site}}(u) S \text{ev}_{w(u)}(Y) R_{\text{site}}(u). \quad (7)$$

Canonical site substitution occurs only in the external coefficients and in  $\text{ev}_{w(u)}$ . All original determining and absolute integer variables are still summed. On an exactly marked orbit, any nonzero determining cutoff forces the unique canonical tuple. Hence both payoffs agree exactly with the originals, including their zero cases.

A common deep coefficient lattice, further deepened for the rational right factors, makes (7) well defined. On bounded group representatives, the left map and  $S$  are bounded; the right map can be reduced to a bounded right-coset representative determined by its denominator and coefficient residues. The triangular inverse in the representation argument then bounds all contributing integer tuples and derivatives. Bounded-range integer evaluation has controlled Lipschitz norm. This gives the required Lipschitz bound, independently of the unrestricted coefficients of  $C$ . Triangular uniqueness preserves the cap one.

*The two field meshes.* After extraction, product-column labels are frozen. The only historical short labels that remain variable are integral affine maps from extra inactive rows. Those rows

annihilate the prepared value space. The chart estimate has the form

$$\|z_i\| \leq \exp(\text{poly}(d))K_i, \quad K_i \leq k_i/v,$$

where  $v$  is the chart width. For a short axis, its scale  $k_i$ , the integral offset, and its values throughout the fixed padded chart have preliminary bounds. The factor  $v^{-1}$  is included in that bound. At every later layer all these costs are warm. The fixed determining polynomials therefore turn the finite row-and-offset menu into a finite menu of entire functions  $u \mapsto w(u)$ .

Transport historical real labels by Lemma 2.2, keeping each higher residual intact until physical reassembly. They are affine expressions in the current normalized spatial coordinates and named higher residuals. They are bounded on the same padded chart, even when they leave a formerly selected unit interval. Retain the original smooth cutoffs and score slice; no new moving sharp indicator is inserted. Outside this fixed padded chart use the zero extension from the representation construction of  $\mathcal{M}$ .

At one site, mesh bounded left values, full-weight coefficients, bounded integer evaluations, right cosets, and inherited cutoff parameters. For a mesh uniform over all sites, mesh the bounded left *coefficient vectors*, the finite whole-function short menu, rational coefficient residues, and the inherited field parameters. The canonical site functions are fixed throughout. These give the distinct point-value and full-field meshes required by Definition 8.1 of  $\mathcal{M}$ .

First pigeonhole the finite tuple of actual historical denominators and cover moduli, then take their product and a fixed deepening. Its logarithm is a sum of warm budgets. Taking the least common multiple of all integers below an upper bound is unnecessary. After product-column freezing, historical raw labels are affine in raw spatial and determining integers and the remaining extra inactive labels. Later integer sections may have arbitrarily large coefficients; they change only the rational numerators, which are recorded modulo the common deep grid. The same divisibility gives the long-label periods. Selection pays a warm loss in the fraction of available productive paths; every retained path keeps its original full score and slice.

These history costs may depend on the creation-layer width and are only preliminary there. They are not inserted into that layer's warm relative width loss. The boundary estimates of  $\mathcal{M}$ , pp. 71–72, use the cap one for arbitrary unsigned payoffs; the averaging on pp. 73–74 uses the total selection measure; and domain localization on pp. 74–75 does not approximate the polynomial payoff. Its Lipschitz bounds apply to the normalized pivot and interval tests. In particular, the micro proof's preliminary short threshold is distinct from the warm small-axis cutoff used for exact domain localization. Larger axes use their normalized affine interval tests.

*Extraction and the final degree.* For a macro comparison, reassemble the physical formula. On its scored locus, replace canonical integers in external coefficients by their corresponding integer summation variables. Exact marking and the original cutoffs identify the expressions. Current auxiliary labels now enter only through this polynomial formula, and both expected representative identities apply before the stochastic constants are frozen. Section 9 of  $\mathcal{M}$  uses these identities, the original slice and masks, cap one, exact determining and inactive equations, and a degree ceiling. It does not require bounds on coefficients or spatial derivatives of the returned physical polynomial. Its extraction and localization therefore apply with the original payoffs.

Every recorded historical-label substitution has final spatial/determining weighted degree at most  $s$ . For a monomial of original absolute-input weight  $h'$  and other charge  $d$ , the joint invariant gives  $h' + d \leq h$ , where  $h \leq s$  is its absolute destination weight. Give each absolute slot the temporary weight  $(s + 1)h$ . Simultaneous substitution gives

$$(s + 1)h' + sd \leq (s + 1)(h' + d) \leq (s + 1)h. \quad (8)$$

This reweights absolute inputs and destinations together. Consequently the physical ceiling is  $s(s + 1)$ .

These temporary weights are used for extraction and final compression. The next middle orbit still uses the original weights, with determining slots and historical integer variables formal; its outer factors again enter the fields. This closes the induction without repeated degree inflation. No family or partner coordinate becomes an absolute slot, so the root rank is still  $d_0$ , independent of old numerical widths.

Finally,  $m$  historical variables of total charge at most  $s$  have at most

$$\binom{m+s}{s} \leq (m+1)^s$$

monomials per scalar coordinate. There are at most  $s$  macro layers and  $s$  internal degree reductions per return. Rational systems, coefficient arrays, and finite meshes are handled simultaneously. The added operations have polynomial sequential depth in  $s$ ; their quantitative bounds are included in Section 4.  $\square$

### 3 The charged active return

We prove Lemma 2.1 by supplementing the degree induction of Theorem 8.2 of  $\mathcal{M}$ . The point is to retain total degree when short variables become coefficient variables, while each recursive child still has strictly smaller positive degree.

*Proof of Lemma 2.1. Charged families and positive normalization.* Record rational coefficient spaces  $A_{d,\ell}$ , decreasing in both indices, with

$$[A_{d,\ell}, A_{e,m}] \subset A_{d+e,\ell+m}, \quad A_{0,\ell} = A_{1,\ell}, \quad A_{r+1,\ell} = 0,$$

and one finite total-charge cutoff in  $\ell$ . Initially  $A_{d,\ell} = \mathfrak{g}_{\max(d,\ell)}$ , with  $\mathfrak{g}_0 = \mathfrak{g}_1$ . Promoting a short block to a dummy  $w$  replaces the array by

$$A'_{d,\ell} = \left\{ \sum_{\beta} w^{\beta} a_{\beta} : a_{\beta} \in A_{d,\ell+|\beta|} \right\}. \quad (9)$$

The positive layers are  $\exp A'_{i,i}$ , and the degree-zero group is  $\exp A'_{0,0}$ . Quotients use images of these spaces, and refinements use the actual intersections before taking images. Every raw and normalized occurrence of a promoted label is changed simultaneously, including those in marks and ancestor factors.

Perform the common-site marking correction before fast-algebra selection. A coefficientwise ordinary filtered section commutes with evaluation at the actual short label, so the inserted correction evaluates to the identity. Normalize the fixed marked constant by a bounded left factor and a rational right factor. The normalized mark takes values in the positive marked group. Let  $c = g(0, 0; w)$  be the constant family element after this marked normalization. The remaining constant coset modulo the positive group can now be removed in the kernel: choose a positive-layer lift  $c_1$  of its marked image, and split the difference  $n_0 = cc_1^{-1} = e_N \gamma_N$  into a bounded kernel element and a kernel lattice element. Replace the family orbit  $g$  by  $e_N^{-1} g \gamma_N^{-1}$ , retaining both factors for restoration. This lies in the positive group because its constant coset does. These are family-group operations, so all charge bounds are retained. A deep source lattice with the prescribed image handles the marked rational factor; its numerator need not be bounded.

*Refiltration with the old degree retained.* The degree-zero action is made compatible with the refiltration by adjoining to the detected fast algebra the rational graded ideal

$$S_i = \text{image} \left( A'_{i+1,i} \longrightarrow A'_{i,i} / A'_{i+1,i+1} \right).$$

The bracket rule makes degree-zero conjugation trivial modulo  $S_i$ , and  $S_r = 0$ . Thus saturation preserves the detected top and its injective marked projection, while the resulting refiltration is normalized by the degree-zero group.

Write  $G_i$  for the old positive layers and  $H_i$  for the refiltration from Lemma 7.6 of  $\mathcal{M}$ . The prescribed full factors preserve old joint degree: a grade- $i$  symbol correction is lifted into  $G_i$ ; the off-space substitution preserves weighted degree; and the constant adjustment sees only old grade one, the higher short degrees already lying in  $G_2 \subset H_1$ . A pure-long grade- $i$  symbol is independent of the short variables, since every coefficient of positive short degree lies in old  $G_{i+1}$ ; its kernel lift therefore preserves total degree. Filtered multiplication and conjugation preserve these bounds. The refiltered input therefore satisfies

$$[v^\alpha z^\beta] \log h \in \mathfrak{h}_{|\alpha|} \cap \mathfrak{g}_{|\alpha|+|\beta|}, \quad \mathfrak{h}_0 = \mathfrak{h}_1. \quad (10)$$

Here  $v$  denotes the long variables and  $z$  the current short variables.

Since  $G_{i+1} \subset H_i \subset G_i$ , the layers

$$H_1 \supset G_2 \supset H_2 \supset G_3 \supset \cdots$$

form one rational flag, with its strong image in the marked group. A rational section adapted to this flag preserves the required intersections. It gives a fixed exact lift  $p(X, z)$  of the refiltered mark with

$$[X^\gamma z^\beta] \log p \in \mathfrak{h}_d \cap \mathfrak{g}_{d+|\beta|}, \quad d = \text{wt } \gamma.$$

Only this interleaved local flag is used; historical charge is already encoded in the coefficient-family group.

*One lower-degree child.* Let  $q : H \rightarrow Q = H/H_r$ , and let  $N$  be the actual kernel of the marking on  $H$ . Top injectivity identifies  $N$  with the kernel of the quotient marking via  $q|_N$ . Promote the current short block to a new dummy  $w$  and form

$$\begin{aligned} \mathcal{Q}_i &= \left\{ \sum_{\beta} w^\beta a_\beta : a_\beta \in q(\mathfrak{h}_i \cap \mathfrak{g}_{i+|\beta|}) \right\}, \\ \mathcal{N}_i &= \left\{ \sum_{\beta} w^\beta n_\beta : n_\beta \in q(\mathfrak{n} \cap \mathfrak{h}_i \cap \mathfrak{g}_{i+|\beta|}) \right\}. \end{aligned}$$

At index zero use  $\mathfrak{h}_0 = \mathfrak{h}_1$  and  $\mathfrak{g}_0 = \mathfrak{g}_1$ . These are images of the displayed intersections. The bracket relation  $[\mathcal{Q}_i, \mathcal{N}_j] \subset \mathcal{N}_{i+j}$  defines the split rational family  $\mathcal{N} \rtimes \mathcal{Q}$ , marked by its second coordinate. Its canonical section is  $a \mapsto (1, a)$ . The map  $(n, a) \mapsto (na, a)$  embeds it into two copies of the quotient family, so it is a rational nilpotent group. Both degree- $r$  spaces vanish, so its positive group has degree at most  $r - 1$ .

The local pair

$$(qh(qp)^{-1}, qp)$$

has the required kernel coefficients by (10). Correct its marking to the common site with the canonical section, and apply the degree-zero normalization above. The resulting input is jointly filtered in all actual variables: substituting a weight- $d$  site monomial can produce actual degree  $a \leq d$ , and a dummy degree  $b$  is charged in the old layer  $\mathfrak{g}_{d+b} \subset \mathfrak{g}_{a+b}$ . Apply the induction to this single child, leaving  $w$  unevaluated.

Restore the child's preliminary degree-zero factors, so its exact second coordinate is  $qp(X, w)$ . Multiplying by the inverse of its canonical marked section isolates the kernel family. Lift this family through  $(q|_N)^{-1}$ , multiply by  $p(X, w)$ , and evaluate  $w = z$ . Its degree- $a$ , dummy-degree- $b$  coefficients lift into  $\mathfrak{n} \cap \mathfrak{h}_a \cap \mathfrak{g}_{a+b}$ . Thus reconstruction restores the original joint degree exactly, including synchronized normalized copies. A common deep coefficient lattice evaluates inside the

pointwise lattice at every integer  $z$ . Fields pull back through the homomorphism  $(n, a) \mapsto na$ ; pullback preserves supremum meshes, and bounded short evaluation has controlled Lipschitz cost. No metric bound on the particular marked lift  $p$  is used.

*Ordered factors and the original score.* Keep the child return in the full split marked group. The joint-fiber reconstruction  $T$  is equivariant for actual kernel factors:

$$T((e, 1)c(r, 1)) = \tilde{e}T(c)\tilde{r}.$$

Indeed the two sides have the same quotient and exact mark, and the joint embedding is injective. The lifts  $\tilde{e}, \tilde{r}$  are rational and filtered. Collect them through the prescribed bounded left and rational right factors of  $\mathcal{M}$ . For the local notation  $K_E\rho(E_F)$  and  $\rho(R_F)K_R$ , this gives

$$E = K_E \text{Ad}_{\rho(E_F)}(\tilde{e}), \quad C = \rho(E_F)T(c)\rho(R_F), \quad R = \text{Ad}_{\rho(R_F)^{-1}}(\tilde{r})K_R.$$

The left conjugation has bounded normalized coefficients, and the right one has controlled rational denominator. The middle has no long-label argument.

If  $L$  is the marking restorer of  $\mathcal{M}$ , use  $E(LC)R$ . It has the required marking everywhere and agrees with  $LECR$  on the full exact-inactive forecasting locus, where  $L = 1$ . That is the locus of both original payoffs and their representative identities. At an earlier central quotient, lift the outer factors separately in the actual kernel and lift the middle separately. Any difference from the original selected lift is removed by the independent central Haar average. Restore the initial degree-zero factors and project away partners. The marking and joint degree are exact, and the expected payoffs, original slice, and additive analytic errors are those of  $\mathcal{M}$ .

At degree zero the scalar-comparison base case applies. Every positive step has one strictly smaller-degree child. This proves the lemma. All additional operations are simultaneous rational linear algebra, bounded-degree coefficient constructions, and one split product per step.  $\square$

## 4 The external inverse budget

For degree  $r$ , let  $I_r$  be an admissible exponent such that a  $U^{r+1}$ -norm lower bound  $\delta$  for a one-bounded  $f : [N] \rightarrow \mathbb{C}$  supplies a correlating degree- $r$  nilsequence with

$$d + \log M + \log(2 + \|F\|_{\text{Lip}}) + \log(1/\text{corr}) \leq K_r(2 + \log(1/\delta))^{I_r}. \quad (11)$$

During constructions of vector-valued nilcharacters we include the *logarithm* of their output dimension in the budget. It need not be the dimension of a nilmanifold.

### 4.1 Uniform coordinate and counting operations

**Lemma 4.1** (Bulk operations). *Let  $q$  bound the nilpotence degree, filtration nesting length, total multidegrees, and orbit-variable count in an operation of [3, 2]. Put  $X = 2 + d + \Lambda$ , where  $d$  includes all growing manifold-coordinate and Lie-generator counts. Vector-output dimensions and numbers of Fourier alternatives and charts enter only through their logarithms in  $\Lambda$ . The latter also includes the other logarithmic numerical data just specified. The nonrecursive rational, coordinate, metric, Fourier, and universal-group operations used below admit bounds*

$$X_{\text{out}} \leq K_q X^{H_q}, \quad H_q \leq \exp(q^{O(1)}).$$

*Joint linear algebra and the fixed-grid lattice constructions are treated as single operations. The bound does not charge the sizes of unrestricted polynomial-orbit coefficients.*

*Proof.* For an  $m$ -by- $n$  rational matrix of logarithmic height  $L$ , clear the denominators jointly, at logarithmic cost at most  $mnL$ . Maximal minors and Cramer's rule bound kernels, images,

sections, and intersections by a fixed polynomial in  $m + n$  times  $1 + L$ . Stack simultaneous conditions before elimination. A basis of a nested flag is selected from its spanning lists, followed by one change of basis; its height need not be recursively powered once per dimension.

The BCH formula has bracket length at most  $q$ . A coordinate expansion uses at most  $K_q(1 + d)^{O(q)}$  indexed terms. A triangular inverse through at most  $q$  nesting blocks has degree at most  $q^q$ . Thus even expanding every needed coefficient array gives at most

$$K_q(1 + d)^{\exp(O(q \log(q+1)))}$$

entries. Denominator clearing, differentiation, coefficient extraction, and rational elimination on these arrays have the asserted bound. The same coordinate polynomials compare the metrics on bounded fundamental representatives. Their Lipschitz constants have logarithm bounded by  $K_q(1 + d)^{\exp(q^{O(1)})}(1 + \Lambda)$ . Here one must not power a budget along an arbitrarily long path in the group metric. There is a direct bound for its entire product. Write its increments as  $\exp(Y_i)$  and put  $S = \sum_i \|Y_i\|$ . In the tensor algebra truncated at degree  $q$ , the degree- $a$  coefficient mass of the product of these exponentials, and then of its logarithm, is bounded by  $K_q S^a$ . Evaluating the Lie words gives

$$\left\| \log \prod_i \exp(Y_i) \right\| \leq K_q M^{K_q} (1 + d)^{O(q)} \sum_{a=1}^q S^a.$$

Coordinate conversion bounds  $S$  polynomially in the total coordinate path length with an order-controlled degree. Taking the infimum over paths yields the metric-to-coordinate bound at once. Polynomial difference factorization gives the converse bound and weak left-translation estimates. Thus the number of path segments does not enter the degree of a budget polynomial.

For a rational subgroup, define its Lie algebra by one integer matrix and use the common weak-lattice denominator. A bounded lattice point close enough to this rational subspace must lie in it: a nonzero defining linear form has magnitude at least the reciprocal common denominator. Its separation threshold has a logarithmic bound of the same type. Below that threshold the quotient displacement belongs to the subgroup lattice; above it, use bounded diameter divided by the threshold. This proves the subgroup and quotient metric comparisons without dimension descent.

Two apparently longer lattice loops deserve explicit mention. In [3, Lemma B.11], one first fixes a weak-lattice integer  $Q_0$ , and all the resulting basis coefficients have height at most  $Q_0^2$ . In [2, Lemma 10.4], the common divisibility constants and weak-lattice inclusions are fixed before collecting the words. Arbitrarily many collections therefore do not create arbitrarily many budget powers. Rational powers and periodicity use the same fixed nesting blocks and joint coefficient denominators [3, Lemmas B.13–B.14].

A free nilpotent group on  $m$  generators of positive weights and allowed total weight at most  $q$  has a spanning word list of size at most

$$\sum_{a=1}^q 4^a m^a \leq K_q(1 + m)^q.$$

Relations within a fixed word type change order-dependent coefficients; the growing generator count is explicitly retained in this dimension estimate. The rational operations just proved then apply to the universal constructions in Section 10 and Lemma C.6 of [2].

On a  $d$ -dimensional torus, Fourier truncation at Lipschitz constant  $L$  and error  $\eta$  has logarithmic term count  $O(d^2(\log(2+d) + \log(2+L) + \log(1/\eta)))$ . Finite chart partitions have the corresponding logarithmic count. The common-function step in [2, Lemma 6.2] selects a member of such a fixed partition; it does not enumerate a net of all Lipschitz functions. Structure-constant selection costs a polynomial in  $d$  and  $\log M$ . Tensor products add logarithmic output dimensions. Neither these coordinates nor all Fourier alternatives are made into separate nilmanifold factors.

The partition construction in [2, Lemma B.3] has one child on the quotient by the last degree layer. At a fixed coordinate precision  $\eta$ , the fibre chart counts have total logarithm  $O(d \log(C/\eta))$ . Local metric comparisons and the product rule give their Lipschitz bounds; there are at most  $q$  degree levels. A bump of width  $\eta$  is assigned the valid derivative bound  $O(\eta^{-1})$ . Nilcharacter embedding (Lemma B.4) uses this partition once. The shear construction (the deferred proof of Lemma 11.3) has rational word and clock-module bases. Clock derivatives copy or kill a word rather than lengthening it; its function is an old function times cutoff factors. The product rule charges their number and the logarithm of the seam width. No iteration in that width is required.

This also accounts for the rational quotient and pullback helpers 3.10 and 10.5–10.12 of [2], and its degree-rank factorization in Appendix B. Their homomorphisms are linear in first-kind coordinates; kernels, images, ideals, and the required changes of basis are the simultaneous matrix operations above. The coefficient and scalar helpers in [3, Appendix A] use the same operations. Its scalar Weyl, Vinogradov, and extrapolation estimates have losses of the form  $\eta^{-C_q}$  or  $Q^{C_q}$ ; after logarithms their unspecified scalar constants are coefficients. They are not powers of the growing logarithmic budget. Its refined bracket estimates have explicit factors polynomial in the growing dimensions after logarithms. The Fourier step in C.6 of [2] can use the torus truncation of [3, Lemma A.5] in the controlled local coordinates just described.

The additive-combinatorial primitive [2, Lemma A.1] already has absolute polynomial-logarithmic powers, including its growing dimension. It can therefore be inserted as one bulk operation. Its internal character-selection count need not be interpreted as a sequence of new generic budget powers. Combining this finite list of coordinate and counting constructions proves the lemma.  $\square$

## 4.2 Affine numerical dependence in equidistribution

**Lemma 4.2** (The complete equidistribution operation). *Let  $q$  bound the degree, central step, filtration nesting and number of orbit variables in Leng’s Theorem 4. Let  $d$  be the input Lie dimension and let  $\Lambda \geq 2$  contain the logarithmic structural, Lipschitz, rationality and inverse-correlation data, together with relative scale losses and requested side cutoffs. It does not include the unrestricted orbit coefficients or  $\log N$ . The logarithmic output data and sufficient side requirements are at most*

$$K_q(1+d)^{E_q}\Lambda, \quad E_q \leq \exp(q^{O(1)}).$$

*The groups supplied to its recursive calls have dimension at most  $K_q d$ , independently of  $\Lambda$ . Thus the complete theorem is one combined-budget operation of degree  $\exp(q^{O(1)})$ .*

*Proof.* We refine Lemma 4.1 by separating  $d$  and  $\Lambda$ . Joint denominator clearing, minors and rational sections have logarithmic height linear in the original logarithmic height. Coordinate arrays have at most  $K_q(1+d)^{\exp(q^{O(1)})}$  entries, and their coefficients are sums of products of order-bounded length. The entire-product metric estimate and the fixed-lattice separation argument used above retain this linear dependence. Fourier selection costs the logarithm of its number of alternatives, which is a structural polynomial times the logarithmic Lipschitz and error data.

The scalar Weyl, Vinogradov and extrapolation estimates in [3, Appendix A] use order-only powers of the numerical heights, errors and retained-scale reciprocals. The refined bracket bounds in Section 3 have factors of the form  $(\delta/(d^{O(d)}M))^{-O(1)}$ ; after logarithms, their  $d \log d$  term is a structural polynomial. Their vector versions have  $O_q(d)$  coordinates. Simultaneous character factorization, rationality pigeonholing and the period and progression operations in Appendices A–B have the same affine dependence. Consequently a nonrecursive stage has the bound

$$\Lambda_{\text{out}} \leq K_q(1+d)^{H_q}\Lambda, \quad H_q \leq \exp(q^{O(1)}).$$

Its coefficient arrays are charged in  $H_q$ . They are not the Lie dimension passed to the next theorem call: the actual groups are squares, subgroups and quotients. A square has dimension at most  $2d$ , and at most  $O(q)$  squares occur on a recursive path.

The two calls in Case 2 of [3, Section 4] are sequential, so they must both be retained. Write  $e(a, b)$  for the structural power at polynomial degree  $a$  and central step  $b$ , and  $e_1(a, b)$  for Case 1. Case 1 has one smaller-central-step child and the simultaneous recovery operations just listed, hence

$$e_1(a, b) \leq e(a, b - 1) + CH_q.$$

Case 2 first invokes the theorem on a normalized square of degree  $a - 1$ . It then refilters a subgroup of the original group, selects a rational progression, and invokes Case 1 with the updated numerical data. The latter input still has Lie dimension at most the original  $d$ . Its two numerical bounds compose as

$$\Lambda_1 \leq K_q(1 + d)^{e(a-1, b) + CH_q} \Lambda, \quad \Lambda_2 \leq K_q(1 + d)^{e_1(a, b)} \Lambda_1.$$

Thus the structural powers add:

$$e(a, b) \leq e(a - 1, b) + e(a, b - 1) + C'H_q.$$

The abelian base has power  $O(H_q)$ . A Pascal majorant gives  $e(a, b) \leq C''H_q 2^{a+b+2}$  for  $a, b \leq q$ , proving the result. In particular, no power of  $\Lambda$  has been composed once per node of this branching calculation.  $\square$

### 4.3 The non-inverse dependency graph

**Lemma 4.3** (One order of the original inverse proof). *With its two lower-order inverse calls removed, an order- $r$  construction in [2] has logarithmic-budget degree*

$$F_r \leq \exp(r^{O(1)}).$$

*Proof.* The operation count is by recursive indices, not by the number of statements with fixed-order notation. One may take a common order parameter  $q = O((r + 1)^2)$ . The ordinary degree-rank chains have length  $O(r^2)$ . The multidegree constructions use at most  $O(r)$  variables with coordinate degrees at most  $O(r)$ , hence total degree  $O(r^2)$ . In particular, Lemma C.6 is used only on these downsets, not on an arbitrary downset from its general statement.

Here is a concrete inventory for Appendix C. Write  $R(\mathbf{a})$  for the coordinate downset below  $\mathbf{a}$ . The two-variable splittings in Lemma 7.5, Lemma 12.1, and Section 12, Step 5, use  $R(0, r - 1)$  and  $R(r - 1, r - 2)$ . The five-variable splitting at the end of Lemma 7.5 uses

$$R(r - 1, 0, 0, 0, 0), \quad R(r - 2, r - 1, r - 1, r - 1, r - 1).$$

The three-variable errors in Section 12, Steps 1–2, use unions of rectangles with coordinate degrees at most  $r - 1$ ; a union of four is handled by three binary splittings. Polarization uses the multilinear box  $\{0, 1\}^r$  and at most  $r + 1$  actual variables. All these downsets therefore have total degree  $O(r^2)$  and at most  $\exp(O(r \log(r + 1)))$  indices.

| Appendix C operation                            | Dependent construction                                                                                                           |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Transitivity and pullback, C.1–C.2              | Products, Taylor substitution, and at most one subgroup basis; all multi-indices are handled together.                           |
| Cancellation, translation, dilation, roots, C.3 | Two-copy subgroup, one quotient, and boundedly many basis operations. The root case calls the dilation case once.                |
| Additivity, C.4                                 | One three-copy subgroup, a Taylor identity, one basis, and one quotient.                                                         |
| Multilinearization, C.5                         | One root call, at most $2^q$ independent subgroup bases, a direct sum, one B.4 call, and a final graph subgroup and quotient.    |
| Splitting, C.6                                  | Bases for the original multiindex layers, at most $q$ total-degree stages, one universal group, one B.3 call, and two quotients. |

The representation step left implicit in C.6 has a direct bulk implementation. With  $g(0) = 1$ , write its logarithm in monomials with coefficients in the prescribed multiindex layers. At total degree  $a$ , express all such coefficients in the fixed layer bases and multiply by the inverse product of the corresponding one-parameter factors. Since lower degrees have already vanished, BCH commutators affect only degrees greater than  $a$ . At most  $q$  stages finish the representation. Its arbitrary real orbit coefficients are not complexity inputs. The resulting universal group has at most  $|J|d$  generators and dimension at most  $K_q(1 + |J|d)^q = K_r(1 + d)^q$ , as required by the bulk lemma. There is no recursive C.5 or C.6 invocation. The subgroup bases indexed by subsets or multiindices come from the same original data, before taking a common product.

We now count actual dependency depth, taking the complete Leng equidistribution theorem as one operation by Lemma 4.2. Its two sequential branches have already been paid inside that lemma. The central-target reductions in [2, Theorem 5.4] take at most  $q$  successive steps, and Corollary 5.5 invokes that result once. The partition in Lemma B.3 and the converse estimate in Lemma B.5 each have one smaller-degree child. Neither makes a new forward inverse call.

The seven stages of the sunflower construction, Lemma 8.3, are sequential. Lemma 9.1 first uses that output and then makes a new Lemma 7.5 and Corollary 5.5 invocation, followed by its fixed linear algebra and the absolute-degree approximate-homomorphism primitive. These successive uses are retained. Lemma 6.3 is then applied at most  $r - 1$  times, each time to the preceding rank's output. The universal-group and extraction constructions of Sections 10–11 have at most  $q$  total-degree stages and a fixed acyclic helper list. Section 12 has a fixed list of main stages using these results.

The large independent families require a different count. All subset bases in C.5, all multiindex bases in C.6, and all factors in a polarization expansion are constructed from the data prescribed before their stage. If a stage has  $T$  factors and target error  $\eta$ , use error  $\eta/(10T)$  for every factor, expand the whole expression, and make its correlation selection once. Products of expansions use the product of their coefficient masses. They do not pass a newly selected correlation threshold from one factor to the next. The number of such factors is order-dependent, at most  $\exp(q^{O(1)})$ . Their direct product dimension is charged, but this multiplies the input dimension by an order-dependent coefficient; it does not apply a new budget power once per factor.

Multilinear expansion can be performed variable by variable with at most  $O(q^2)$  addition levels; within a level the factors are handled as above. The C.6 representation has its  $q$  true sequential total-degree cancellations. Powers such as  $r!$  and  $r^r$  affect rational coefficients and tensor output dimensions, not the filtration degree or the number of inverse calls. The actual multidegrees remain those in the inventory above.

A conservative bound on the resulting dependency depth is  $C_0(q + 1)^6$ , where  $C_0$  is absolute. This covers the  $O(q^3)$  sunflower/linearization blocks, their rank iteration, and the remaining fixed nesting of degree and quotient operations. Taking  $H_q \leq \exp(q^{O(1)})$  to include the full equidistribution operation, the complete body degree satisfies

$$F_r \leq (2H_q)^{C_0(q+1)^6}, \quad \log F_r \leq r^{O(1)}.$$

This estimates the actual constructions, including their parallel assemblies, rather than interpreting an unspecified  $O_r(1)$ .  $\square$

#### 4.4 Integration of supplied derivative correlations

We replace the last forward inverse call in Section 12 of [2] by a recursion which retains its already constructed lower-degree correlators. The following elementary calculation supplies the precise recursive input. Set  $\Delta_t^+ A(n) = A(n + t) \overline{A(n)}$ .

**Lemma 4.4** (Mixed correlations supply derivative correlations). *Let  $A$  be one-bounded on  $[L]$ , extended by zero, let  $B$  be one-bounded on  $[2L]$ , and let  $\psi_h$ ,  $h \in [L]$ , be one-bounded sequences.*

Suppose

$$\frac{1}{L} \sum_{h=1}^L \left| \frac{1}{L} \sum_{n=1}^L A(n) B(n+h) \psi_h(n) \right| \geq \rho.$$

If  $L \geq 4\rho^{-2}$ , there are sequences  $\Xi_t$ ,  $1 \leq t < L$ , each a product of two translated or conjugated members of the original family, or zero, such that

$$\frac{1}{L} \sum_{t=1}^{L-1} \left| \frac{1}{L} \sum_{n \in \mathbb{Z}} \Delta_t^+ A(n) \Xi_t(n) \right| \geq \rho^2/8.$$

At least  $\rho^2 L/16$  positive shifts have correlation at least  $\rho^2/16$ . If the  $\psi_h$  are degree- $a$  nilsequences on manifolds of dimension at most  $d$ , the  $\Xi_t$  have the same degree and dimension at most  $2d$ . No regularity in the shift parameter is required.

*Proof.* Choose unit phases  $c_h$  to make the sum of the inner correlations real and at least  $\rho L^2$ . With  $m = n + h$ , Cauchy–Schwarz removes  $B$  and gives

$$\rho^2 L^4 \leq 2L \sum_{m=1}^{2L} \left| \sum_{h=1}^L c_h A(m-h) \psi_h(m-h) \right|^2.$$

Expand the square and average in its second shift  $u$ . For some  $u \in [L]$ , taking absolute values termwise yields

$$\sum_{t \in u-[L]} \left| \sum_{n \in \mathbb{Z}} \Delta_t^+ A(n) \psi_{u-t}(n+t) \overline{\psi_u(n)} \right| \geq \rho^2 L^2/2.$$

The zero-shift term is at most  $L$ , so the nonzero terms total at least  $\rho^2 L^2/4$ . For  $t = -v < 0$ , substitute  $n = m + v$  and conjugate; the witness for  $\Delta_v^+ A(m)$  becomes  $\overline{\psi_{u+v}(m)} \psi_u(m+v)$ . At most two terms fold to one positive shift. Keep the larger correlation, losing at most a factor two. The asserted average and the density of shifts follow. The zero extension belongs to  $A$ , not to a nilsequence witness. Product groups and fixed orbit translations prove the last assertion without estimating any orbit coefficient.  $\square$

For the omitted small case  $L < 4\rho^{-2}$ , the mixed hypothesis implies  $\mathbb{E}|A|^2 \geq \rho^2$ . Parseval on  $\mathbb{Z}/L\mathbb{Z}$  gives a linear phase with correlation at least  $\rho/\sqrt{L} \geq \rho^2/2$ . Its logarithmic budget has the required bound. This also handles small-interval alternatives below.

**Proposition 4.5** (Supplied derivative integration). *For  $r \geq 2$ , suppose a one-bounded  $f : [N] \rightarrow \mathbb{C}$ , extended by zero, has one-bounded degree- $(r-1)$  nilsequences  $\theta_h$ ,  $h \in [N]$ , satisfying*

$$\mathbb{E}_{h \in [N]} \left| \mathbb{E}_{n \in [N]} \Delta_h^+ f(n) \theta_h(n) \right| \geq \rho.$$

*Let  $X \geq 2$  bound their dimensions, all logarithmic numerical data, and  $\log(1/\rho)$ . Then  $f$  correlates with a degree- $r$  nilsequence whose dimension and logarithmic numerical and inverse-correlation budget is at most  $K_r X^{J_r}$ , where*

$$J_r \leq \exp(r^{O(1)}).$$

*The witnesses may use different manifolds and arbitrary polynomial orbits, subject only to the common budget.*

*Proof.* Let  $F_r$  be the complete non-inverse body degree from Lemma 4.3, enlarged by an absolute power to include the elementary lemma and the final products. For  $r = 2$ , the fixed-degree converse estimate of [2, Lemma B.5] and the Gowers derivative identity give a  $U^3$  lower bound with inverse logarithm  $X^{O(1)}$ . The established fixed-order  $U^3$  inverse theorem supplies a finite absolute  $J_2$ .

For  $r \geq 3$ , threshold the supplied correlations and begin [2, Lemma 6.2] immediately after its initial inverse call. All subsequent steps accept the witnesses as data: normalization at zero, selection of common structure constants and filtration, selection from the fixed function partition, vertical Fourier selection, and nilcharacter embedding. In particular, no net of all Lipschitz functions is used. They produce precisely the degree-rank correlation structure of Definition 6.1 of that paper.

Apply its Lemma 6.3 through the rank descent and then its Section 12 symmetrization. Those statements require this correlation structure, not a fresh Gowers-norm hypothesis. At the end of Section 12, before its last use of the converse estimate, the scalar coordinate selection gives

$$\mathbb{E}_h |\mathbb{E}_n A(n)B(n+h)\psi_h(n)| \geq \rho', \quad \deg \psi_h \leq r-2.$$

Divide by the controlled supremum bounds. The interval and shift ranges can be padded by zeros as in the original proof. Apply Lemma 4.4 to obtain supplied derivative correlations for  $A$ , and apply the present assertion at degree  $r-1$ . Its output is a degree- $(r-1)$  correlator for  $A$ , exactly the object required in the last paragraph of the original proof. Multiplication by its already known degree- $r$  factor, substitution, and residue-class recombination finish the argument. The earlier converse invocation in Section 12, Step 3, is retained and is part of  $F_r$ .

If an intermediate size condition fails,  $\log N$  is bounded by the current body budget. The supplied correlation gives a lower bound on  $\mathbb{E}|f|^2$ ; Parseval then supplies a linear-phase correlator with logarithmic cost  $O(\log N + \log(1/\rho))$ . Thus this alternative is within the same budget. The only recursive integration call in the main alternative is the one at  $r-1$ . Charging the product of the before- and after-recursion body degrees inside  $F_r$ , a sufficient recurrence is

$$J_r \leq F_r(1 + J_{r-1}).$$

Since  $\log F_r \leq r^{O(1)}$ , summation of logarithms proves the asserted bound.  $\square$

**Proposition 4.6** (Single-exponential inverse budget). *The exponents in (11) may be chosen with*

$$I_r \leq \exp(r^{O(1)}).$$

*Proof.* The  $U^2$  Fourier argument gives  $I_1 = 1$ , and the fixed order  $r = 2$  has a finite absolute exponent. For  $r \geq 3$ , the derivative identity and the degree- $(r-1)$  inverse theorem give the input to Proposition 4.5, with budget  $K_r(2 + \log(1/\delta))^{I_{r-1}}$ . The derivative norm and shift-density thresholds are powers  $\delta^{C_r}$ , whose logarithms introduce coefficients only. Therefore

$$I_r \leq (1 + I_{r-1})(1 + J_r).$$

Taking logarithms and summing the bounds on  $\log(1 + J_r)$  gives  $\log(1 + I_r) \leq r^{O(1)}$ . Both stages of the original argument have been accounted for; the second now uses a proved integration interface with the supplied lower-degree correlators.  $\square$

## 5 Relative detection by cube degree

Throughout Sections 5–6, let  $\mathcal{H}_D \leq \exp(D^{O(1)})$  be a common degree allowance for the elementary coordinate, mask, and completion operations. Section 6 enlarges it to include the source’s nonrecursive operations. It contains no forward inverse or relative-detector invocation.

### 5.1 Effective cube completion

The following statement concerns ordinary integer cubes. It is uniform in the coefficients of the polynomial orbit.

**Lemma 5.1** (Deletion of one cube vertex). *Let  $q \geq 2$ , let  $G_0 = G_1 = G$ ,  $G_q = 1$ , and let  $Y = G/\Gamma$  have dimension  $d$  and an adapted Mal'cev basis of complexity  $M \geq 2$ . Let  $F : Y \rightarrow \mathbb{C}$  be one-bounded and  $L$ -Lipschitz, where  $L \geq 1$ , and let  $0 < \eta < 1/2$ . For any vertex  $i \in \{0, 1\}^q$ , there are coefficients  $c_a$  and one-bounded Lipschitz functions  $F_{a,\omega} : Y \rightarrow \mathbb{C}$ , for  $\omega \neq i$ , such that every adapted polynomial map  $g : \mathbb{Z}^n \rightarrow G$  satisfies*

$$\left| F(g(x + i \cdot h)\Gamma) - \sum_a c_a \prod_{\omega \neq i} F_{a,\omega}(g(x + \omega \cdot h)\Gamma) \right| \leq \eta \quad (12)$$

for all  $x, h_1, \dots, h_q \in \mathbb{Z}^n$ . The functions and coefficients are independent of  $g$ . With  $X = 2 + d + \log M + \log L + \log(1/\eta)$ , they may be chosen so that

$$\log\left(2 + \sum_a |c_a|\right) + \max_{a,\omega} \log(2 + \text{Lip}(F_{a,\omega})) \leq K_q X^{\exp(q^C)}. \quad (13)$$

Each individual factor uses the original nilmanifold of dimension  $d$  and has degree at most  $q - 1$ .

*Proof.* Reflections and permutations of cube coordinates reduce the assertion to deletion of the all-ones vertex. Identify vertices with subsets of  $[q]$ . For  $S \subseteq [q]$ , write  $[a]_S$  for the element of  $G^{\{0,1\}^q}$  equal to  $a \in G_{|S|}$  at vertices containing  $S$ , and equal to the identity elsewhere. The upper-face cube group  $\mathcal{C}$  consists of the ordered products

$$c = \prod_{S \subsetneq [q]} [a_S]_S, \quad a_S \in G_{|S|},$$

where the ordering extends inclusion. The full face is absent because  $G_q = 1$ . Face commutators satisfy  $[[a]_S, [b]_T] = [[a, b]]_{S \cup T}$ ; commutator closure of the filtration therefore gives this product description. Successive proper vertex values recover all  $a_S$ , so deletion defines an injective homomorphism

$$\Pi : \mathcal{C} \longrightarrow P := G^{2^q-1}$$

onto a closed rational subgroup  $H$ . Filtered polynomial maps send integer cubes into  $\mathcal{C}$ : their successive face coefficients lie in  $G_{|S|}$ , by the iterated-difference definition of a polynomial map, and the last coefficient lies in  $G_q = 1$ .

*The full lattice.* If all proper vertex values of  $c$  belong to  $\Gamma$ , triangular recovery gives  $a_S \in \Gamma \cap G_{|S|}$ ; their product at the missing vertex also belongs to  $\Gamma$ . Consequently

$$\Pi(\mathcal{C} \cap \Gamma^{\{0,1\}^q}) = H \cap \Gamma^{2^q-1}. \quad (14)$$

Thus deletion is injective on the nilmanifold quotients as well. There is an explicit adapted basis: if  $X_j$  has filtration weight  $w_j$ , take  $\mathbf{1}_{[S]} \otimes X_j$ ,  $|S| \leq w_j$ ,  $S \subsetneq [q]$ , ordered first by the original Mal'cev index  $j$ . Its dimension is at most  $2^q d$ . Copies of the same  $X_j$  commute, and at a vertex  $T$  the  $j$ -th second-kind coordinate is exactly

$$z_j(T) = \sum_{\substack{S \subseteq T \\ |S| \leq w_j}} t_{j,S}.$$

Integer Boolean Möbius inversion on proper vertices shows that its integer-coordinate lattice is precisely  $\mathcal{C} \cap \Gamma^{\{0,1\}^q}$ . Brackets go to later original Mal'cev indices; their coefficients are inherited from the original structure constants.

*Effective inverse and external lattice gap.* Put  $A_q = \exp(q^{C_0})$ , increasing the absolute  $C_0$  below, and let

$$R = \exp\left[K_q(1 + d)^{A_q}(1 + \log M)\right].$$

We claim that completion on the image in  $Y^{2^q-1}$  is  $R$ -Lipschitz, after enlarging  $K_q, C_0$ . In first-kind coordinates,  $\log H$  is defined, separately in each original coordinate  $j$ , by the integral equations saying that its proper-vertex values form a Boolean polynomial of degree at most  $w_j$ . Each equation has coefficients  $0, \pm 1$  and at most  $2^q$  terms. Möbius inversion also gives an integral linear inverse for  $\Pi$  on this subspace.

All BCH and coordinate-change expressions have bracket length at most  $q - 1$ . Listing the bracket words, clearing the at most  $d^3$  input structure-constant denominators, and bounding these polynomials and their derivatives on bounded coordinate boxes gives bounds of the form  $R$ . Their number and coordinate degrees are at most  $K_q(1 + d)^{Aq}$ ; this uses no repeated substitution indexed by the dimension. In particular, for every  $\gamma \in \Gamma^{2^q-1}$ , its first-kind coordinates have a common denominator at most  $R$ , independently of the magnitudes of its integer second-kind coordinates. If  $\gamma \notin H$ , one defining integral equation for  $\log H$  therefore has absolute value at least  $R^{-1}$  at  $\log \gamma$ .

Represent two points of the cube nilmanifold by  $a, b$  in its coordinate fundamental box. Their images, and  $\Pi(b)^{-1}\Pi(a)$ , have bounded coordinates of the preceding form. If the quotient images are  $\epsilon$ -close, some ambient lattice element  $\gamma$  makes  $\Pi(a)$  and  $\Pi(b)\gamma$  at most  $2\epsilon$  apart. Coordinate and bounded-translation estimates imply

$$\|\log \gamma - \log(\Pi(b)^{-1}\Pi(a))\| \leq R\epsilon.$$

For  $\epsilon$  below an explicitly bounded reciprocal power of  $R$ , the external lattice gap forces  $\gamma \in H$ . Equation (14) then makes this an actual cube-lattice correction. The linear inverse and coordinate estimates bound the intrinsic quotient distance by  $R^{O(1)}\epsilon$ . For larger  $\epsilon$ , use the bounded diameter of the cube fundamental box and divide by that explicit threshold. Evaluation at the missing vertex is a homomorphism with a  $0, 1$  Lie matrix, so it has the same type of metric bound. Absorbing these fixed powers proves the claimed global completion bound. This argument supplies the effective inverse; qualitative compactness alone would not do so.

*Uniform tensor interpolation.* Let  $m = 2^q - 1$ , and let  $K \subseteq Y^m$  be the compact image of deletion. The function  $F \circ \text{completion}$  on  $K$  is one-bounded and  $LR$ -Lipschitz in the maximum product metric. Extend its real and imaginary parts by the McShane formula and project radially onto the closed unit disk. The resulting function  $\tilde{F} : Y^m \rightarrow \mathbb{C}$  is one-bounded and  $CLR$ -Lipschitz.

Choose a  $\delta/2$ -net  $z_1, \dots, z_J$  in  $Y$ , where  $\delta = \eta/(10CLR)$ . Mal'cev coordinate boxes give

$$\log J \leq K_q(1 + d)^{Aq}(1 + \log M + \log(1/\delta)).$$

Set  $b_j(y) = \max(0, 1 - d_Y(y, z_j)/\delta)$  and  $\phi_j = b_j/\sum_l b_l$ . The denominator is at least  $1/2$ , so  $0 \leq \phi_j \leq 1$ ,  $\sum_j \phi_j = 1$ , and  $\text{Lip}(\phi_j) \leq CJ/\delta$ . For  $a = (a_1, \dots, a_m)$ , take  $c_a = \tilde{F}(z_{a_1}, \dots, z_{a_m})$  and use  $\phi_{a_j}$  at the  $j$ -th retained vertex. The tensor partition is a convex combination of unit-disk coefficients, so the *entire sum* is one-bounded on all of  $Y^m$ . A nonzero term has its center tuple within  $\delta$  of the input tuple. Its uniform approximation error is therefore at most  $\eta$ . Also  $|c_a| \leq 1$  and  $\sum_a |c_a| \leq J^m$ . The displayed net estimate and  $\log R$  prove (13), after increasing the absolute exponent. Restriction to  $K$  proves (12).  $\square$

For cyclic variables evaluated in integer representatives, the lemma is used only after fixing a sufficiently short box tuple and its wrap pattern. The resulting virtual integer cube has vertex representatives  $u_\omega = y_\omega - a_\omega P$ , with fixed integer wrap vectors  $a_\omega$ . Apply the lemma to  $g'(y) = g(y - a_i P)$ . A retained factor is then evaluated at  $g(u_\omega + (a_\omega - a_i)P)$ , a fixed translation of its polynomial orbit. Such a translation changes neither the nilmanifold nor the Lipschitz function. Box selection and boundary errors are accounted for separately; no dependence on  $P$  or on orbit coefficients enters Lemma 5.1.

## 5.2 One inverse call at each cube order

**Proposition 5.2** (Relative detection by degree descent). *The conclusion of  $\mathcal{M}$ , Lemma 5.2, admits a budget exponent  $R_q$  satisfying, after an absolute enlargement of the nonrecursive allowance  $\mathcal{H}_D$ ,*

$$R_1 \leq \mathcal{H}_D, \quad R_q \leq \mathcal{H}_D(I_{q-1} + 1)(R_{q-1} + 1) \quad (2 \leq q \leq D). \quad (15)$$

*The output remains warm. Required errors in the bounded-grid and spatial marginal estimates have logarithmic budgets of the same type and are charged to the cold record. In particular,  $R_q \leq \exp(D^{O(1)})$ .*

*Proof.* For  $q = 1$ , independent group sites give the detector mean, so the niltest is constant. Assume henceforth that  $q \geq 2$ . Use a product of cyclic groups  $H = \prod_a \mathbb{Z}/P_a\mathbb{Z}$ , where  $P_a$  is prime and

$$4qH_a \leq P_a \leq 8qH_a.$$

Thus  $P_a > q$ , and box/group normalizations cost at most  $K_q^n$ . The proof of  $\mathcal{M}$ , Lemma 5.1, uses representative wrap cells and leading symbols, so this change of comparable moduli preserves its bounded-grid and marginal conclusions.

Put the detector at vertex 0. Retain the inverse-free densification on pp. 37–38 of  $\mathcal{M}$ : it makes every other factor one-bounded, leaving  $|F_0| \leq B_{\text{cap}}\nu$  and a correlation  $\delta_0$  with

$$\log(1/\delta_0) \leq K_q(1 + \log B_{\text{cap}} + \log(1/\delta)).$$

The repeated Cauchy–Schwarz and interpolation steps here act affinely on these logarithms. Their number changes  $K_q$ .

Choose the opposite vertex  $i = \mathbf{1}$ . Its conditional dual  $D_i$  obeys  $|D_i| \leq 2B_{\text{cap}}$ , by pairwise uniformity of distinct group-cube sites. The current correlation gives  $\mathbb{E}|D_i|^2 \geq \delta_0^2$ . Insert  $D_i$  at that vertex and use Gowers–Cauchy–Schwarz and  $\|F_0\|_{U^q} \leq 2B_{\text{cap}}$ , the latter following from the majorant moments. Consequently

$$\|D_i/(2B_{\text{cap}})\|_{U^q} \geq \delta_0^2/(4B_{\text{cap}}^2).$$

One ordinary inverse call replaces vertex  $i$  by a one-bounded degree- $(q-1)$  nilsequence  $T$ , retaining correlation  $\rho$ . Its dimension, logarithmic complexity, and  $\log(1/\rho)$  have degree  $\mathcal{H}_D(I_{q-1} + 1)$  in the original growing budget, after enlarging  $\mathcal{H}_D$ .

*Localization and the missing mask.* Partition every representative coordinate at every vertex into discrete intervals of representative diameter less than  $P_a/(100q)$ , allowing singletons, with boundary cutoffs placed halfway between adjacent grid points. There are at most  $K_q^n$  tuples of cells. One retains correlation  $\rho_1 \geq \rho/K_q^n$ . Keep their indicators as one-site factors. For proper vertices define the virtual integer cube

$$y_\omega = u_0 + \sum_a \omega_a(u_{e_a} - u_0).$$

The short cells fix  $y_\omega - u_\omega = a_\omega P$  for every proper  $\omega$ ; their proper-subcube faces determine these offsets without the missing vertex. The selected full tuple also fixes  $y_i - u_i = a_i P$ , with all offsets bounded in terms of  $q$ .

Smooth the  $i$ -th indicator from inside and multiply it into  $T$ . This is essential: an arbitrary indicator left at  $i$  would prevent deleting that vertex. Its error is bounded by

$$\mathbb{E}|F_0(u_0)|\mathbf{1}_{\partial I_i}(u_i) \leq 2B_{\text{cap}}\mathbb{P}(u_i \in \partial I_i),$$

again by pairwise uniformity. A relative transition width  $\theta$  with  $B_{\text{cap}}n\theta \ll \rho_1$  suffices. For sub-grid widths the half-grid endpoints permit exact agreement on the discrete grid. Realize

the selector on circles of period  $K_q P_a$ , large enough that its only supported lift in the possible range of  $y_i - a_i P$  is the selected real cell. It adds  $n$  dimensions and has logarithmic Lipschitz cost  $O_q(\log(1/\theta))$ ; no  $\log P_a$  cost is needed.

Apply Lemma 5.1 to this modified nilsequence on the common orbit  $t \mapsto g(t - a_i P)$ . On the other vertices its arguments are the fixed translates  $u_\omega + (a_\omega - a_i)P$ , which do not change orbit complexity. The identity remains valid on the entire region defined by the other vertex masks: if the selector contributes,  $y_i - a_i P$  is in its chosen real cell and is congruent to  $u_i$ , so it equals  $u_i$ ; conversely the selected cells fix the required offset whenever  $u_i$  lies there. Otherwise both masked values vanish. Thus deleting  $i$  introduces no extra scoring region, and uses no periodicity of  $g$ .

Request uniform completion error  $\eta \ll \rho_1/B_{\text{cap}}$ . Its cube-average error is at most  $2B_{\text{cap}}\eta$ . If  $M$  is the coefficient mass of the completion expansion, one term retains correlation  $\sigma \geq \rho_1/(2M)$ . All its proper-vertex factors are one-bounded; only the detector is sparse. Every new one-site nilsequence has the dimension of the single inverse-output group, including the selector circles. The possibly large number of summands contributes  $\log M$ , not a product of that many nilmanifolds.

*A lower-order cube norm.* For  $t = 1, \dots, q-1$ , group the remaining nonzero vertices by Hamming weight  $t$ . The direction  $v_t = (t, -1, \dots, -1)$  annihilates their linear forms, whereas its value at the detector form is  $t$ , invertible on  $H$ . Cauchy-Schwarz along these  $q-1$  directions therefore gives, for arbitrary  $F$  and one-bounded  $f_\omega$ ,

$$\left| \mathbb{E}_{x,h} F(x) \prod_{\omega \neq 0, \mathbf{1}} f_\omega(x + \omega \cdot h) \right| \leq \|F\|_{U^{q-1}(H)}.$$

Indeed the final detector arguments are  $x + \sum_t \epsilon_t t h'_t$ ; changing the independent variables  $th'_t$  gives the usual cube norm. No independence of the directions  $v_t$  is required. Other factors are removed by their pointwise bounds, so no new linear-forms condition on  $\nu$  is used.

Write the resulting detector as  $F_0 T_0 \mathbf{1}_{I_0}$ . Its  $U^{q-1}$ -norm is at least  $\sigma$ . Apply the smaller-order relative statement to its diagonal cube, of correlation at least  $\sigma^{2^{q-1}}$ . In the original notation retain  $J_0$  and replace only the arbitrary bounded input  $P_0$  by  $P_0 T_0 \mathbf{1}_{I_0}$ . The returned degree- $(q-2)$  niltest multiplied by  $T_0$  is an ordinary degree- $(q-1)$  niltest.

Finally smooth the root mask at the already determined correlation scale. Lemma 5.1's spatial/chart marginal bound gives the error

$$\mathbb{E}|F_0| |\mathbf{1}_{I_0} - W_{I_0}| \leq B_{\text{cap}} \mathbb{E} \nu \mathbf{1}_{\partial I_0} \leq B_{\text{cap}} (O_q(n\theta) + \epsilon_{\text{cold}}).$$

Half-grid endpoints again cover sub-grid widths. This uses no pointwise bound for  $\nu$ . Put this degree-one mask into the final niltest. The case  $q = 1$  is the direct mean argument.

To return to the atom in  $\mathcal{M}$ , Theorem 3.4, use its unchanged chart-measure normalization on pp. 39–40. In that notation,  $\chi = 1$  on the support of  $J_0$  and  $a_\chi/m_j$  is warm, so the returned atom is  $J_0$  times the degree- $(q-1)$  niltest. All output choices above are warm; only the marginal and grid comparison errors enter the late cold requests.

There is one ordinary inverse call, one completion operation, and one smaller-order relative call. Products and all required errors have the same ordered budgets, proving (15). Taking logarithms of that recurrence and using Proposition 4.6 gives  $\log(R_q + 1) \leq D^{O(1)}$ , as claimed.  $\square$

## 6 The source-local numerical schedule

**Proposition 6.1** (Local polynomial degrees). *For the construction above and the imported analytic framework, the fresh absolute rank is at most  $K_k p^F$ , independently of the incoming path dimension. A common degree  $B$  bounds all nested structural path-dimension polynomials, the*

width-increment polynomials after eliminating the intermediate numerical records, and the root logarithmic cost and side-requirement polynomials. One may choose

$$1 \leq F \leq B, \quad 2 \leq B \leq \exp(D^{O(1)}). \quad (16)$$

At weight  $i$ , the width increment may use only incoming  $Q_j$  with  $j > i$ . Root requirements may use all widths.

*Proof.* There are two separate estimates: the degree of a nonrecursive operation in its growing inputs, and the number of successive substitutions of such estimates. We record both.

*Nonrecursive operations.* Include every growing dimension and coordinate count in a common budget  $x$ . Joint rational operations and bounded-degree coefficient arrays satisfy Lemma 4.1, with an order parameter polynomial in  $D$ . A degree- $D$  polynomial in  $n$  growing variables has at most  $(n+1)^D$  coefficients. In contrast, the Boolean jet of degree  $h$  on a cube of order  $q$  has at most

$$j(h, q) = \sum_{a=0}^{\min(h, q)} \binom{q}{a} \leq 2^q$$

coordinates per axis. Its fixed order-dependent size is not a new inverse-theorem order or a new determining weight.

The remaining elementary analytic bounds also have uniform polynomial degrees. The lattice-extraction bound in  $\mathcal{M}$ , Lemma B.2, has the explicit form

$$C_* = \delta^{-1} (C(n+m)R)^{C(n+m)};$$

its logarithm is polynomial in the dimensions and the logarithmic requests. The chart matrices in Appendix F have logarithmic norm  $O(d^3)$ . For the real sublevel estimate in Appendix G, the reciprocal exponent is  $Nd$ ; the one-axis minor uses at most  $j(h, q)h(q+1)$  variables and degree at most  $j(h, q)(h-1)$ . Its order-dependence is therefore at most  $\exp(D^{O(1)})$ . The simultaneous Jacobian is block diagonal. Applying the estimates axiswise and summing errors adds the logarithm of the number of axes; it does not multiply the polynomial degrees of all minors. The many-multiples induction in Lemma G.3 has a recurrence  $C_a \leq K(4C_{a-1} + 10)$ , with an absolute  $K$ , and hence exponential dependence on the distinguished monomial degree.

The modular estimates in Appendix H use differencing loss  $2^{1-h}$  and a block count of size  $K_h(C+m+1)$ . Those block counts are structural and are fixed before numerical accuracy. Its exceptional-prime argument tests individual prime powers and product witnesses. The large integer depending only on  $h$  enters as an additive constant in a logarithmic height, not as a power of  $x$ . Likewise flag truncation in Section 7.4 has  $\log Q_d \leq 2 \log T \log Z / \log 2$ , a product of two growing logarithms. No step takes the product of all moduli below an accuracy cutoff. If a growing Fourier derivative order is needed, choose the initially free smooth cutoffs as standard Gevrey bumps with derivative constants bounded by  $C^{a+1}(a!)^2$ . Their logarithms are polynomial in  $a$ ; scaling adds  $a$  times the scale logarithm.

The selected-plane construction has an explicit structural dimension bound as well. For  $\ell$  parameter directions and output degrees  $d_i \leq D$ , its equation and derivative counts are

$$J = \sum_i \left[ \binom{\ell + d_i}{d_i} - 1 \right], \quad J' = \sum_i \binom{\ell + d_i - 1}{d_i - 1}.$$

The moment and derivative failure estimates in the proof of Lemma 6.1, pp. 48–49, are satisfied by a free dimension  $K_D(\ell + J + J' + 1 + \log(1/c_*))$ . Thus it is polynomial in the structural inputs with degree  $O(D)$ . Hensel lifting has an exact normalized count at every prime-power depth and does not update a global budget at each depth.

For the degree-one comparison, the dependence on the fixed slack must also be separated from the polynomial degree. In  $\mathcal{M}$ , pp. 101–107, the moments are  $O_\epsilon(x)$  and  $\log M = O_\epsilon(x)$ . The largest regularity amplification  $M^{2q}$  has logarithm  $O_\epsilon(x^2)$ , so one can choose the regularity error  $\exp[-K_\epsilon x^2]$ . The dependent random choice then has inverse-density logarithm  $O_\epsilon(x^2)$ . Formula (C.4) gives per-round rank increase  $O_\epsilon(x^8)$  and logarithmic width loss  $O_\epsilon(x^2 + \log(2+d))$ . After  $O_\epsilon(x)$  rounds, rank, width logarithm, and mass logarithm have powers at most 9, 3, 12, respectively. Smoothing the Bohr tests fits within  $K_\epsilon(2+x)^{64}$ . The long concentration-peeling loop is paid by the telescoping potential (C.8), with the allowed successor shapes fixed beforehand. Thus changing the slack at later degrees changes coefficients only. Positive counting in Appendix D has at most  $s$  axis compressions and, before those compositions, an initial logarithmic cap budget of degree  $s+2$ .

These observations, together with the joint-array operations of Sections 2–3, give a nonrecursive degree bound

$$\mathcal{H}_D \leq \exp(D^{O(1)}). \quad (17)$$

Fourier modes, finite menus, and empirical model terms contribute their full logarithmic counts. Selected pivot partners contribute their structural product-group dimension. This dimension is an input to the next operation; it is not dropped from the bound.

*Parallel major refinements with sharp output records.* The remaining apparent exponential chains are the tensor restoration in Lemma 4.7 and the degree tuples in backward rank transfer. Their full call trees have exponentially many nodes, but only degree-decreasing edges carry a new cold inverse output.

To see this, use the actual arithmetic conclusion of  $\mathcal{M}$ , Lemma 4.6, rather than its looser polynomial-logarithmic summary: if  $d'$  clears the substituted rational polynomial, its denominator divides  $d'D_0$ . This holds for all tensor entries at once with a common denominator. Primitive extension and coefficient-one extraction preserve that denominator. A local denominator may initially vary with the productive path, so one should not assert divisibility by a single initial integer without first selecting it. Instead, for the logarithmic bound  $r_t$  of the current common denominator, a safe update is

$$r_{t+1} \leq A_D(n)(1 + r_t + \log D_0 + \Lambda_{>j}). \quad (18)$$

Here  $A_D$  is a structural polynomial of degree at most  $\exp(D^{O(1)})$ , and  $\Lambda_{>j}$  contains permitted higher-layer lattice data. In the native primitive coordinates the coefficient of  $r_t$  can be taken to depend only on  $D$ ; the displayed bound safely allows joint coordinate changes. Taking a product of two denominators adds their logarithms. There is no new power of  $r_t$ .

The real conclusion of Lemma 4.7 is equally sharp: the coefficient norm  $B_{\text{slow}}$  is multiplied by a dimension-controlled factor, independently of the I.1 output complexity. The factors in (4.6) are canceled by the final normalized  $z^\alpha$  multiplication. Thus its logarithmic norm has an affine update of the same kind as (18). All scalar entries are processed jointly. The number  $L$  of degree stages or degree tuples depends only on  $D$ , not on the structural dimension. Even allowing  $L \leq \exp(D^{O(1)})$ , iterating these affine updates gives a structural power at most

$$L \deg A_D \leq \exp(D^{O(1)}).$$

It does not give  $(\deg A_D)^L$  as a power of the precision variables. Fixed-order multipliers may be absorbed into  $K_D$ .

In Lemma 4.7, pp. 32–33, the parent first calls I.1 and fixes  $K'$ , its threshold, and its preliminary height data. Every extra-sheet-degree child is then invoked *starting with this same  $K'$* . A child may internally restrict to  $K''$ , but extends its resulting identity back to its input  $K'$ . The next sibling receives the sharp slow/rational identity and its common denominator, not that child's  $K''$ , cold height, or inverse-output complexity. The forward bound and denominator bound hold

on all the original productive path substitutions, so subtraction does not require passing a newly selected internal path family to the next sibling. Forecast the sharp affine envelope for every sibling, use that common input bound, and take the maximum of their late requirements.

A parent-to-child edge really does carry the parent's newly frozen sides and  $K'$  description, and is counted. Its weighted degree decreases by at least one. Put  $I = \max_{r \leq D+2} I_r$  and  $R = \max_{q \leq D} R_q$ . One I.1 block has a degree bound

$$V_D \leq [2\mathcal{H}_D(I + R + 1)]^{D^{C_0}}$$

for an absolute  $C_0$ . Here the block consists of the original translation-group realization, the F.2 Property 2 and Appendix H detector, the B.1 degree descent, the I.3–I.4 algebraic reconstruction, and the joint scalar components. Their genuine dependency depth is polynomial in  $D$ ; they do not call Lemma 4.7 or backward rank transfer. Thus this definition has no dependence on the refinement being estimated. If  $E_d$  denotes the resulting degree at weighted degree  $d$ , a sufficient recurrence is therefore

$$E_0 \leq \mathcal{H}_D, \quad E_d \leq \mathcal{H}_D V_D (1 + \max_{e < d} E_e), \quad E_d \leq (2\mathcal{H}_D V_D)^{O(d)}. \quad (19)$$

The full tensor tree remains paid through its joint counts and affine coefficient envelope; it is not treated as a serial list of new I.1 precision powers.

At an ancestor in pp. 33–35, at most  $(D + 1)^D$  degree tuples are processed. Exact separation identifies the new slow tensor with the old local slow tensor, and the new rational tensor with the old rational tensor, *before* multiplying by deviation or raw monomials. This prevents the unrestricted coefficients of later determining maps from entering the next residual budget. The sharp records again satisfy (18); higher widths enter through at most  $D \sum_{i > j} \log((w_i \epsilon_i)^{-1})$ . No cold I.1 output from one tuple is needed for the next.

Degree-zero pairs can require a productive-fraction selection. For  $r$  tensor entries, denominator bound  $Q$ , and bounded difference radius  $C$ , their number has logarithm at most

$$O(r^2 \log(2 + Q) + r \log(2 + C))$$

in fixed higher-layer primitive coordinates. A translate of the bounded region has the same bound. These losses add over the order-bounded tuple list; they depend on its sharp affine envelope, not on earlier I.1 output complexities. Forecast their total once and use a common input fraction for every tuple. Internal further selections remain inside the already pruned modular-good branch; they do not require a new  $D_0$  based on the smaller fraction. These selections prove existence of the split and, as stated at the end of p. 35 of  $\mathcal{M}$ , are not propagated as ancestor coefficient or denominator costs. All late side requirements are nevertheless retained, by taking their maximum and passing the full cold record to lower layers. There are at most  $D$  genuine ancestor passages.

*The remaining schedule.* Appendix B has one smaller-degree child and at most  $D$  grade corrections. Appendix C has two successive modeling phases and one smaller-degree comparison; its second phase is uniform over the first list. Appendix E has at most  $D$  fixed-map budget levels. Appendices G–H have degree-index product-block estimates and the explicit fourteen-stage schedule. I.6 has  $D$  degree descents and  $D$  grade corrections, with pivot partners requested together. I.15 has one smaller-old-weight child. Extraction and the added geometry have at most polynomially many layer and degree descents. Rank cuts use joint row systems and additive costs, as in Section 4.6 and I.11–I.12 of  $\mathcal{M}$ .

The four-stage numerical order of  $\mathcal{M}$ , Section 10.3, is unchanged. The full record  $U_j$  includes current preliminary and late costs and passes them to lower layers. It does not feed them back into that layer's width increment. Eliminating these records takes  $D$  further descending substitutions. Structural path-variable counts and precision-dependent model dimensions are included in their respective inputs throughout.

After flattening the fixed helper graph, the genuine nonlinear depth is  $D^{O(1)}$ . The order-bounded enumerations just analyzed contribute joint structural degrees and affine losses within  $\mathcal{H}_D \leq \exp(D^{O(1)})$ . Proposition 5.2 replaces the old cube-vertex chain. A sufficient common bound is

$$F, B \leq [2\mathcal{H}_D(I + R + 1)]^{D^{O(1)}} \leq \exp(D^{O(1)}). \quad (20)$$

This includes all nested structural path-dimension polynomials. The fresh-rank independence is the bottom absolute rule,  $\mathcal{M}$ , Lemma 2.2, obtained from additive relative rank with zero old rank. Width increments retain the strict higher-layer dependence of the original contracts. This proves the proposition.  $\square$

## 7 Finite-horizon density iteration

We give the complete final calculation, including a separate allowance for the structural path-variable counts.

*Proof of Theorem 1.1.* Let  $F, B$  be the exponents in Proposition 6.1, enlarged so that  $1 \leq F \leq B$  and  $B \geq 2$ . The bottom rule creates at most  $K_k p^F$  fresh absolute coordinates per round, independently of the incoming path dimension. All dimensions are forecast before any numerical precision is chosen.

*Dimensions and cuts.* In one preparation round the original dimension ledger of  $\mathcal{M}$ , Proposition 10.2, is

$$H_D = d_D, \quad H_i = d_i + H_{i+1}^2, \quad d_i^{\text{out}} \leq d_i + d_0 + H_{i+1}^2,$$

with  $H_{D+1} = 0$ . A cut at  $i$  decreases  $\dim W_i$ ; only exports from above can replenish this dimension. Thus  $H_i$  bounds its cut count within the round, including restarted tests. Over  $T \leq K_k p$  rounds starting from no slots, sufficient exponents for the dimension envelopes are

$$a_D = F + 1, \quad a_i = 1 + 2a_{i+1}, \quad a_i = (F + 2)2^{D-i} - 1.$$

Indeed the additional factor  $p$  pays the number of rounds; descending induction bounds both the incoming dimensions and the within-round  $H_i$ . Therefore

$$a_0 = 2^D(F + 3)$$

bounds all coordinate counts and the cumulative number of cuts as  $K_k p^{a_0}$ . The structural path-column constructions are polynomials in these counts of degree at most  $B$ , by the local budget proposition. We consequently use the common structural forecast

$$d_* \leq K_k p^a, \quad a = B 2^D(F + 3). \quad (21)$$

The extra factor  $B$  explicitly pays for these path variables and coefficient arrays formed from width-independent structural variables. Precision-dependent model and partner dimensions are included in the local numerical operations, not in  $d_*$ . Thus this structural forecast introduces no dependence on a width.

*Widths and root requirements.* The strict higher-layer dependence of the local width polynomials gives

$$Q_i^{\text{out}} \leq Q_i + K_k \left( 2 + p + d_* + \sum_{j>i} Q_j \right)^B.$$

Starting with trivial widths, let  $K_k p^{v_i}$  bound the width logarithm at  $i$  throughout the proposed run. We may take

$$v_D = 1 + Ba, \quad v_i = 1 + Bv_{i+1} \quad (i < D). \quad (22)$$

Here  $v_{i+1} \geq a$ , and each added one again pays the sum over rounds. Hence  $v_1 \leq (2B)^D a$ . Evaluating the root-cost and side-requirement polynomials and summing over the run gives total logarithmic cost at most  $K_k p^A$ , where

$$A = 1 + Bv_1 \leq (2B)^{D+1} a. \quad (23)$$

No current or lower width is substituted into its own increment.

Since  $F \leq B$  and  $\log B \leq D^{O(1)}$ , (21)–(23) imply

$$\begin{aligned} \log A &\leq (D+2) \log(2B) + D \log 2 + \log(F+3) \\ &\leq D^{O(1)} = k^{O(1)}. \end{aligned}$$

Increasing one absolute constant  $C$ , uniformly also for the finitely many small orders, we obtain

$$A \leq \exp(k^C). \quad (24)$$

*The contradiction and inversion.* Use one fixed starting budget  $p$ , enlarging its order-dependent coefficient if necessary so all temporary density thresholds exceed  $e^{-K_k p}$ . Reserve  $T \leq K_k p$  rounds so that the target  $(\alpha/2)(1 + \eta_k)^T$  reaches one, where  $\eta_k > 0$  is the retained multiplicative gain. The cost bound above sustains all rounds whenever

$$\log N \geq K_k (2 + \log(1/\alpha))^A.$$

The original certificate is impossible at target at least one, since its positive payoff is pointwise at most its enlarged negative baseline. This is the finite-horizon argument of  $\mathcal{M}$ , Proposition 10.4; endpoint and side-rounding costs were included above.

Taking  $\alpha = r_k(N)/N > 0$  and using the contrapositive yields

$$\alpha < e^2 \exp\left[-K_k^{-1/A} (\log N)^{1/A}\right].$$

The case  $\alpha = 0$  is immediate. By (24),  $1/A \geq \exp(-k^C)$ . Since  $\log N > 1$  for  $N \geq 3$ , decreasing this exponent and renaming the positive order-dependent constants proves (1).  $\square$

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